

ervised, and reinforcement learning (RL) based methods. We proceed by briefly reviewing these methods and their relation to the problem of knowledge injection.

Supervised Fine-Tuning Supervised fine-tuning (SFT) requires sets of labeled input-output pairs. One of the most common SFT methods is instruction tuning (Wang et al., 2022; Mishra et al., 2021; Ouyang et al., 2022; Taori et al., 2023), which has emerged as one of the most powerful methods to improve model performance. With instruction tuning, the input is a natural language task description, and the output is an example of the desired behavior. Many current state-of-the-art LLMs have gone through instruction tuning after their pre-training phase.

Instruction tuning has been shown to be very effective at improving the overall quality of the model, with a particular emphasis on its zero-shot and reasoning capabilities. However, despite these advantages, instruction tuning does not necessarily teach the model new knowledge (Ouyang et al., 2022; Chung et al., 2022; Mitra et al., 2023; Chia et al., 2023; Zhou et al., 2023). As such, instruction tuning alone is not a viable solution to the knowledge injection problem.

Reinforcement Learning Another form of FT relies on RL or RL-inspired optimization strategies to better align the model after its pre-training phase. A few prominent examples are reinforcement learning from human feedback (RLHF) (OpenAI, 2023; Touvron et al., 2023), direct preference optimization (DPO) (Rafailov et al., 2023), and proximal policy optimization (PPO) (Schulman et al., 2017; Tunstall et al., 2023).

These techniques have been shown to be very useful, especially when used in conjunction with instruction tuning. However, similarly to instruction tuning, these methods focus on the overall quality of the response and its expected behavior and not necessarily on its breadth of knowledge.

Unsupervised Fine-Tuning The final FT strategy we discuss is unsupervised, meaning there are no available labels for the model to learn from. One common unsupervised FT technique is often referred to as *continual pre-training* or *unstructured FT*.

In this method, the FT process is viewed as a direct continuation of the pre-training phase. We start with a saved checkpoint of the original LLM and train it in a causal autoregressive manner, i.e., predicting the next token. One major difference in comparison to actual pre-training is the learning rate. Usually, one would need a much lower learning rate when continuing the pre-training of the model to avoid catastrophic forgetting (Kirkpatrick et al., 2017).

It is well known that LLMs store vast amounts of knowledge during their pre-training phase (Zhou et al., 2023). So, it makes sense to continue this process in order to inject

knowledge into the model. Hence, we use the unsupervised FT approach throughout this work and evaluate its efficacy in enhancing the model’s capacity for learning new information.

3.3. Retrieval Augmented Generation

Retrieval augmented generation (RAG) (Lewis et al., 2020) is a technique that expands LLMs’ capabilities, especially in knowledge-intensive tasks, by using external knowledge sources. While the original formulation involved additional training per task, it has since been demonstrated (Neelakantan et al., 2022) that a pre-trained *embedding* model can achieve improved performance with no additional training involved.

The idea is that given an auxiliary knowledge base and an input query, we use the RAG architecture to find documents within the knowledge base that resemble the input query. These documents are then added to the input query, thus giving the model further context about the subject of the query.

In practice, implementing the suggested architecture is quite straightforward: Given an auxiliary knowledge base $\mathcal{B}_Q$ and a pre-trained embedding model $\mathcal{M}_e$, we create a dense vector representation (embedding) per document $b \in \mathcal{B}_Q$ and store these in a vector store. Upon receiving a new query q , we use its embedding, $\mathcal{M}_e(q)$, to retrieve q ’s top- K closest neighbors, $\mathbf{b}_q = \{b_k\}_1^K$, according to dot-product ranking. We then update q to be $\tilde{q} = \mathbf{b}_q \| q$, where $\|$ denotes string concatenation. Finally, we return $\mathcal{M}(\tilde{q})$ as the model’s output.

4. Knowledge Base Creation

4.1. Task Selection and Rationale

MMLU Benchmark To properly evaluate the capabilities of LLMs on knowledge-intensive tasks, we selected four distinct tasks from the Massively Multilingual Language Understanding Evaluation (MMLU) benchmark (Hendrycks et al., 2021) in the topics of anatomy, astronomy, college biology, college chemistry and prehistory. The chosen tasks were selected based on their emphasis on factual knowledge and the minimal reliance on reasoning. As a heuristic, we opted for tasks where the questions are short and involve no context. In practice we selected four STEM subjects as well as one humanities subject, to ensure the evaluation is not limited to certain fields. Note that prehistory involves questions spanning all non-modern history. This approach aims to enable us to test LLM proficiency in comprehending and manipulating information in isolation from its reasoning processes.

Current Events Task To further isolate LLMs’ abili-

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Table 1. Results for the MMLU datasets described in Section 4.1 in terms of log-likelihood accuracy (Equation (4)).

Task	Model	Base model	Base model + RAG	Fine-tuned	Fine-tuned + RAG
Anatomy (0-shot)	Mistral 7B	0.556	0.681	0.570	0.659
	Llama2 7B	0.393	0.489	0.430	0.489
	Orca2 7B	0.607	0.637	0.600	0.637
Anatomy (5-shot)	Mistral 7B	0.600	0.681	0.622	0.674
	Llama2 7B	0.467	0.563	0.496	0.548
	Orca2 7B	0.570	0.659	0.593	0.674
Astronomy (0-shot)	Mistral 7B	0.625	0.678	0.651	0.697
	Llama2 7B	0.401	0.467	0.487	0.520
	Orca2 7B	0.645	0.750	0.651	0.750
Astronomy (5-shot)	Mistral 7B	0.658	0.724	0.651	0.697
	Llama2 7B	0.401	0.474	0.447	0.520
	Orca2 7B	0.664	0.763	0.664	0.743
College biology (0-shot)	Mistral 7B	0.681	0.757	0.701	0.764
	Llama2 7B	0.438	0.493	0.458	0.465
	Orca2 7B	0.583	0.639	0.604	0.632
College biology (5-shot)	Mistral 7B	0.722	0.778	0.736	0.771
	Llama2 7B	0.451	0.521	0.424	0.479
	Orca2 7B	0.604	0.660	0.625	0.653
College chemistry (0-shot)	Mistral 7B	0.470	0.500	0.490	0.500
	Llama2 7B	0.310	0.380	0.390	0.390
	Orca2 7B	0.370	0.440	0.370	0.390
College chemistry (5-shot)	Mistral 7B	0.470	0.540	0.500	0.500
	Llama2 7B	0.370	0.380	0.360	0.390
	Orca2 7B	0.430	0.470	0.370	0.380
Prehistory (0-shot)	Mistral 7B	0.713	0.750	0.719	0.731
	Llama2 7B	0.448	0.481	0.457	0.478
	Orca2 7B	0.642	0.679	0.673	0.673
Prehistory (5-shot)	Mistral 7B	0.722	0.762	0.725	0.762
	Llama2 7B	0.515	0.531	0.503	0.537
	Orca2 7B	0.664	0.698	0.667	0.694

Table 2. Current events results. Models that were fine-tuned on the original dataset are labeled as *FT-reg*, while those trained on the dataset with multiple paraphrases are labeled as *FT-par*.

	Base model	Base model + RAG	FT-reg	FT-par	FT-reg + RAG	FT-par + RAG
Mistral 7B	0.481	0.875	0.504	0.588	0.810	0.830
Llama2 7B	0.353	0.585	0.219	0.392	0.326	0.520
Orca2 7B	0.456	0.876	0.511	0.566	0.820	0.826

ties to learn new knowledge, we created a task comprising multiple-choice questions about current events. This task includes multiple-choice questions about events that occurred after the cutoff of the various models’ training data. Specifically, we focused on ”current events” from the USA, in the time span of August-November 2023, that are included in

the relevant Wikipedia indexes¹. This method enables us to mostly guarantee that the models have not been exposed to these facts, thus allowing us to directly test knowledge injection capabilities.

¹https://en.wikipedia.org/wiki/Category:2023_events_in_the_United_States_by_month

4.2. Data Collection and Preprocessing

To effectively evaluate the LLMs’ performance on these knowledge-intensive tasks, a comprehensive auxiliary dataset was collected by scraping relevant articles per topic from Wikipedia. The rationale behind selecting Wikipedia as the primary source of knowledge is its broad coverage of relevant topics and its reliability as a repository of crowd-verified knowledge. All articles pertinent to the tasks were retrieved via the official Wikipedia API² by identifying the relevant central page per topic.

Subsequently, a rigorous cleaning process was utilized to transform the data from raw subsections to clean chunks. This step was done with the “wikiextractor” tool (Attardi, 2015). The division into small, clean (e.g., remove HTML, URLs, etc.) chunks was aimed at enhancing the evaluation of the LLMs’ understanding across various knowledge domains and aiding the LLMs in the fine-tuning process.

4.3. Current Events Task Creation

After collecting the relevant chunks from Wikipedia, we created a new multiple-choice dataset with the help of GPT-4 (OpenAI, 2023). First, we removed any small chunks. For each remaining chunk in the corpus, GPT-4 was instructed to create four highly specific, high-quality multiple-choice questions with only one correct answer. By specific, we mean that the question can be answered without knowledge of which context the question refers to and with minimal ambiguity. Next, GPT-4 was asked to select the two most specific of the four. This was followed by a manual evaluation and verification step. In total, this resulted in 910 new questions.

4.4. Paraphrases Generation

After creating the dataset, we utilized GPT-4 to generate augmentations of the dataset. We instructed GPT-4 to provide paraphrased versions of the input data that fully retain the information while being reworded. Each paraphrasing iteration was done with a different seed to ensure variety.

We selected 240 chunks at random for each task and created two paraphrases per chunk. These were set aside to be used as validation sets for hyperparameter tuning. For the current events dataset, we created ten paraphrases for each chunk used in the fine-tuning process described in Section 6.

5. Experiments and Results

Experimental Framework We used the popular LM-Evaluation-Harness (Gao et al., 2021) repository to evalu-

²https://www.mediawiki.org/wiki/API:Main_page

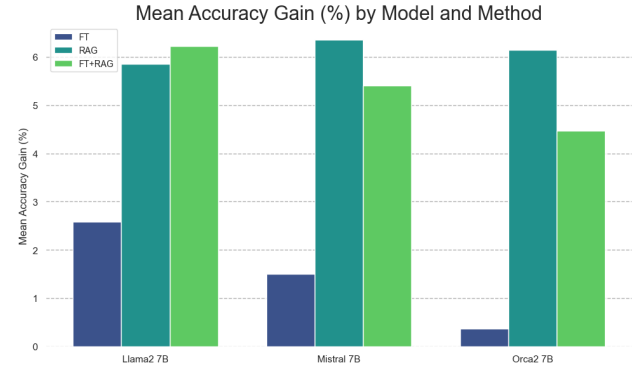


Figure 2. The relative accuracy gain (as explained in Equation (5)) for each knowledge-injection method, averaged (columnwise) across all experiments in Table 1.

ate the performance of LLMs on the selected knowledge-intensive tasks. LM-Evaluation-Harness is a robust benchmarking tool that currently serves as the industry standard for model evaluation and is the basis of the HuggingFace leaderboard³. Leveraging this platform ensured a standardized evaluation framework and allowed consistent comparison across models, methods, and datasets. More importantly, by using the industry standard for evaluation, we could avoid any differences stemming from prompt engineering and formatting issues and replicate the reported baseline results for each model.

Model Selection We chose three models for inference evaluation: Llama2-7B (Touvron et al., 2023), Mistral-7B (Jiang et al., 2023), and Orca2-7B (Mitra et al., 2023). The choice of these models was meant to represent the most popular open-source base models and an instruction-tuned model across various baseline capabilities. Additionally, we selected *bge-large-en* (Xiao et al., 2023) as the embedding model for the RAG component and used FAISS (Johnson et al., 2019) as its vector-store. This embedding model is currently the SOTA of open-source embedding models, according to the HuggingFace MTEB leaderboard⁴.

Configuration Variations Our evaluation included multiple configurations, with a grid-search over them, to allow for more comprehensive benchmarking.

Firstly, we compared the baseline and fine-tuned models and their performance with the RAG component. Secondly, we explored the optimal number of text chunks to add to the context in RAG. Specifically, different values of $K \in \{0, \dots, 5\}$ were employed to analyze the impact on model performance. Finally, we explored 5-shot performance vs. 0-shot.

³https://huggingface.co/spaces/HuggingFaceH4/open_llm_leaderboard

⁴<https://huggingface.co/spaces/mteb/leaderboard>